Comparison of Zero-Shot Ethical Classification With and Without Automatically Generated Consequences

Rafal Rzepka[1] **and Kenji Araki**[1]

[1]Faculty of Information Science and Technology, Hokkaido University
Kita-ku, Kita 14, Nishi 9
Sapporo, Japan
{rzepka,araki}@ist.hokudai.ac.jp

Abstract

It is difficult to recognize triggers which lead to us to making moral decisions in our daily lives and the same is true about artificial moral agents. Systems like Delphi show relatively high accuracy in recognizing potentially problematic acts but a user cannot be sure why a given label was chosen. What if we ask a language model to generate possible consequences of an act and perform sentiment analysis on these consequences? In this paper we report our early findings from a work in progress regarding recognition of contextual changes which might change the moral interpretation of an act and cause a moral agent to perform an action. Our preliminary experiments show that implementing a neural sentence generator does provide input relatively useful for sentiment analysis but the recognition accuracy for a zero-shot classification performed only on the act decreases when longer consequences are generated while it steadily increases in the case of a classic lexicon-based approach. This suggests that richer consequence description might help to improve moral estimation of acts in the future.

Introduction

Although problematic, using wisdom of crowds (WoC) for automatic moral evaluation of acts has been proven to have some potential and the latest neural language models have been improving text classification helpful for predicting various labels for free input (Rzepka and Araki 2017; Schramowski et al. 2019; Ramezani et al. 2021; Hendrycks et al. 2021). Recently, Delphi, a system (Jiang et al. 2021) introduced by Allen Institute for AI has shown that to an extent, a model trained on WWW and ethical examples can access "ethical common sense" knowledge and predict that for example *it's wrong to rob a bank if you are poor* or that *it's inappropriate to wear pajamas to a funeral*. Criticism toward Delphi concentrated on human biases inherited by the model, but there is another problem. Users asking Delphi their own questions[1] can freely add reasons as a context, however there is no explanation generated, only a short phrase as "you shouldn't" or "it's appropriate". For example, when we extend the pajamas at funeral scenario as follows:

> *My father was a funny person. He held pajama parties every year. He even asked all his relatives to wear pajamas to his funeral. But it is not normal. Now he is dead and I am torn. Should I wear pajamas to his funeral?*

Delphi changes its answer to *It's okay*. There is no explanation as "you should prioritize te will of your deceased father" and if we add additional details as "family won't participate if I wear pajamas" or "but it is inappropriate in our Arabic culture", the output remains "it's okay". However, in real life, we anticipate pros and cons of our acts by considering existing reasons and possible consequences, we also use them to explain our choices to others. Another problem, which is due to the characteristics of deep learning, Delphi's users cannot be sure why the system gives a particular answer or why a mistake happened. In the classic approach based on phrase frequencies (Rzepka and Araki 2005) it was possible to refer to the raw corpus data and observe why for example "stealing a car" achieved relatively high positive recognition (e.g. many entries about *Grand Theft Auto* game). With the current neural approaches we are limited to the analysis of the output or the prompts used for querying. However, crowdsourcing possible reasons and consequences for unlimited number of acts is simply impossible. For this reason we decided to perform preliminary experiments with GPT-2 in order to see to what extent automatic generation of consequences may become a viable approach. We have laso utilized zero-shot classification task to see if the generated consequences can improve recognition of ethical and unethical acts.

Related Work

Although Artificial Moral Agents are widely discussed, the number of actually working systems is still rather scarce. Examples of case-based reasoning programs are TRUTH-TELLER (McLaren and Ashley 1995) and SIROCCO (McLaren 2003), the logic approach is also taken by (Pereira and Saptawijaya 2016) or (Anderson and Anderson 2014). When it comes to machine learning, for example (Guarini 2006) used neural nets to teach a machine choose acceptable and unacceptable cases of killing. Until recently, research on machine ethics have been concentrating on theoretical proposals (Greene et al. 2016; Conitzer et al. 2017),

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[1]https://delphi.allenai.org/

label	input act	generated sentence
NEUTRAL	*a woman carries a heavy block with her bare hands*	*It is dangerous to use only the hands and fingertips; use the whole body and move with the body. Good use of the fingertips is a safe and efficient way to exercise.*
NEUTRAL	*mother breaks the washing machine by mishandling it.*	*The reason why my mother broke the washing machine was because my father was trying to put the laundry in the machine. No one knows why my mother broke the washing machine. My father...*
DANGEROUS	*a person kills poisonous insect with bare hands*	*Or, the individual swallows the poison of the organism, which affects its eating habits. Similarly, many people eat the animal's food not as an animal flesh, but as an insect, and ...*
DANGEROUS	*a person with a knife cuts the skin of a security guard*	*Even if you look at the video, the act falls under the category of "attempted murder," which includes "killing a person," "helping a person you attacked," or "injuring a person," but in the video "you shouldn't kill a person"...*
SAFE	*my daughter eats her rice ball with a delicious smile.*	*My daughter was kind of happy to see that, wasn't she? She smiled and ate it :) (that's important! :)) I didn't write about it in today's blog, but I ...*
SAFE	*a player hits a table tennis ball during a game*	*A ping-pong table is not used for table tennis, but is often used in ping-pong halls as a substitute for a table tennis table or racket (see "Racket" for details) Table tennis tables are smaller than table tennis tables*

Table 1: Translated example consequences generated by GPT-2 (50 Japanese characters version). Generated sentences are often unfinished and translated as closely as possible to the original to mirror lack of the naturalness of the output.

but with the advent of machine learning, the task of retrieving and calculating the moral rightness of actions has become relatively popular in recent years (Hoover et al. 2019; Ramezani et al. 2021; Jiang et al. 2021). Also, to address the need for relevant data, several datasets of moral choices have been created (Hendrycks et al. 2021; Emelin et al. 2020; Sap et al. 2020). However, for the real world applications, collecting similar situations and annotating them is costly and time consuming, therefore researchers experiment with neural language models to e.g. classify if an input describes moral or immoral behavior. Language models as BERT (Devlin et al. 2019) or GPT (Radford et al. 2019; Brown et al. 2020) have not only improved scores on NLP tasks, but also gave an opportunity to access the Wisdom of Crowds quickly and deal with context changes to some extent as Delphi example above shows. Schramowski et al. demonstrated that BERT has a better moral compass than previous embedding-based methods (Schramowski et al. 2019) but they have not tested how a language model would decide if the context changes are small. Their experimental setup is also binary (*do*s vs. *don't*s) which raises a question about other dimensions of ethical classification of acts. Their work also suffers from problems found in approaches of (Rzepka and Araki 2017) and (Jentzsch et al. 2019), i.e. actions are simple and these methods are not meant for situations where e.g. a robot needs to decide if a user's action is problematic in the first place. They also do not test other capabilities of language models like mask prediction or language generation which could be helpful in simulating similar cases and guessing possible consequences. By experimenting with automatic ethical evaluation of modified situations, in this paper we test our hypothesis that generating possible consequences of an act would give additional hints for the classifier and provide some level explainability (as one can examine the generated sentences).

Experimental Data

Because there is no dataset dedicated to recognizing ethical nature of an act for Japanese language, we use 1,000 extensions of simple sentences as "to kill a spider" manually rewritten by crowdworkers to sound dangerous or safe (e.g. "to kill a poisonous spider with a bare hand" and "to kill a spider with an insecticide aerosol"). Three labels ('safe', 'neutral' and 'dangerous') are chosen by three different workers and decided by the majority vote. The set is randomly sampled from an over 20,000 sentences from a corpus under development for a different project and it consists of 408 "safe", 330 "dangerous" and 262 "neutral" labels.

Algorithm Description

In order to automatically decide if an input act is ethical, neutral or unethical by using different sets of classification labels, we utilize zero-shot[2] classification with multilingual XLM-R model (Conneau et al. 2020). Finally we use language generation capability of GPT-2 (Radford et al. 2019) to provide additional information on consequences of a given act and perform sentiment classification with both

[2]Historically, zero-shot learning (ZSL) has been meant to train a classifier on one set of labels and then evaluate on a different set of labels that the classifier has never seen before, but in the context of language models it basically means using an existing model trained on a specific task (natural language inference in this case) to solve another task.

	en	ja	pl	zh	Avg
moral, neutral, immoral	69.0	64.1	68.0	64.6	67.0
ethical, neutral, unethical	**70.9**	63.9	69.1	59.9	**68.0**
good, neutral, bad	67.5	63.6	68.7	64.2	66.6
righteous, neutral, wicked	58.5	54.2	68.0	63.9	60.2
virtuous, neutral, sinful	67.7	60.9	62.3	66.9	63.6
voting	58.6	52.0	62.3	54.7	56.9
average	66.7	61.3	**67.2**	63.9	65.1

Table 2: As the XLM-R model is multilingual and labels can work better depending on the language, we performed comparative experiments and decided to utilize English triple "ethical", "neutral", "unethical" as default labels for further experiments with automatic evaluation of consequences.

lexicon and zero-shot approaches. In order to lead the generator to output consequence containing opinion, we add "This is/was very" to the end of every input act (*sore-wa sugoku* in Japanese which is a tense-agnostic expression).

To observe the influence of the generated output size, we generate consequences in three lengths (25, 50, 100 characters). Lexicon-based baseline recognizes words from a combined emotion-related and society-related positive/negative words, therefore except standard adjectives as "good" or "bad" adjectives, it consists also words as "awarded" or "arrested" (99 positive and 165 negative, 264 in total), see (Rzepka and Araki 2017) for details on how the lexicon was created. When the number of positive and negative words in a generated output is equal, the consequences are labeled as neutral.

Experiment and Results

The dataset described above is meant for recognizing dangerous human behavior but we assume that most of the entries contain morally doubtful acts, for example "woman takes somebody's car with keys in the ignition". We use Japanese GPT-2 model[3] to generate three sets of sentence(s) following these acts. For example a short generated "consequence" of the act above is "driver's wife, 43 years old, dies after being hit by a car" (Table 1 shows other examples of generated output). As it can be observed, generated text is often not logically consistent with the input, therefore in near future we plan to use bigger Japanese GPT model which has been published after this paper was accepted. In this stage of research our goal is first to examine if a class (predicted label) of a generated consequence matches the label of the input ("dangerous" in this case). We perform series of experiments and compare lexicon-based approach of (Rzepka and Araki 2017) with the multilingual XLM-R model (Conneau et al. 2020) which is able to classify any textual input with an arbitrary set of labels. After testing five sets of labels in English, Japanese, Polish and Chinese, we decided to use English "ethical", "neutral", "unethical" triple as they achieved the highest agreement with golden labels (see Table 2) .

[3]https://huggingface.co/rinna/japanese-gpt2-medium

	Lexicon	XLM-R ethical	XLM-R positive
len 25	29.7	63.8	**64.4**
len 50	36.7	60.2	64.1
len 100	**43.6**	59.4	62.7

Table 3: Percentage of predicted labels agreeing with the crowdworkers' annotation. "XLM-R positive" stands for a version using "positive","neutral" and "negative" labels to be more comparable with the lexicon-based approach.

As shown in Table 3, none of the models extended with generated consequences have reached 70.9% agreement of XLM-R model without the extension using the same English *ethical-neutral-unethical* labels (see Table 2). However, differently from the zero-shot classification, lexicon's results visibly increase together with the length of generated consequences. This suggests that zero-shot classifiers do not sufficiently deal with longer inputs and older approaches, when equipped with neural text generation methods, might be more useful than expected, especially with their understandability (see the following section) advantage.

Conclusion and Future Work

Our results show that adding more information does not lead to accuracy improvement over a neural zero-shot classification. However, we discovered that more generated context information might be useful also in older approaches relaying on simple counting of lexicon words. Although precision of these classic methods is high, the recall remains low[4]. Adding more context can help to alleviate this problem, but the quality of generation remains the crucial hurdle of this procedure. The more natural context description we acquire, the more precisely we can test existing and future metrics for measuring morality. The more easy-to-understand examples what can happen next we give to a child, the more probable that it will trigger more precise evaluation of an act. Although "matching" is not "understanding", we believe that similar strategy can be utilized in natural language processing. The question is how exemplary and trustful such language model-generations are and this must be examined in the future. The output differs depending on language and different results will be achieved for societies using this language (Rzepka, Li, and Araki 2018), therefore we chose Japanese instead of English for the trials. But this choice causes the biggest problem of the presented experiments, as the lack of proper test data forced us to use a set of sentences meant for recognizing dangerous situations. Because the correlation between morality of an act and the level of its danger is not straightforward (killing a poisonous animal is *dangerous* but is it *unethical*?), we are currently in the process of creating a dataset dedicated to the ethical aspect of changing contexts and the experiments must be repeated to confirm if the findings presented in this paper apply also to annotations intended for ethical classification.

[4]Another way to improve recall is to use the bigger lexicon but our previous experiments have shown that enlarging sentiment dictionaries often cause noise especially when synonyms are utilized.

In the near future we also plan to use masking to see if more specific prompting of neural language models can be used for achieving more precise predictions about consequences, also according to both agent and patient of an act. As we have not yet experimented with reasons of a given act (which can change the classification results), by adding phrases as "the consequence of that is" or "the reason of that is" to the end of the act, we will try to generate sentences describing both consequences and reasons more precisely.

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